CMPS 360 – Data Science Fundamentals

Syllabus and Course Admin



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Outline

- Course introduction
- Grading
- Policies

About the Instructor

- **Dr. Abdelkarim Erradi**
  - CS Associate Professor
  - **Office:** H07 - C309, College of Engineering
  - **Phone:** 4403 4254
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Office hours (to be confirmed):

- **L01 - Sunday 1:30pm to 2:30pm at Male office hours area**
- **L51 - Tuesday 1:30pm to 2:30pm at H07-C309**
- You can talk to me **after** class if you have a quick issue/question
- Best way to contact me is via **MS Teams chat**

Course Learning Outcomes

1. **Explain** fundamental data science concepts, including the data science lifecycle, data quality, exploratory analysis, statistical inference, modeling, and responsible data use

2. **Apply** computational, statistical, and data science techniques to acquire, prepare, analyze, visualize, and interpret real-world data

3. **Design and implement** data science workflows to address data-driven problems, and **evaluate and communicate** their results effectively

Prerequisites

- **CMPS 151 - Programming Concepts** is the formal prerequisite for this course
- **Proficiency in Python**. All assignments and the project will use Python. Check your level with the following tutorials
  - https://docs.python.org/3/tutorial/
  - https://numpy.org/doc/stable/user/absolute_beginners.html
- **Basic Probability and Statistics**. You should be familiar with the basics of probability, distributions, mean, standard deviation, etc.
  - https://people.engr.tamu.edu/guni/csce689/files/prob.pdf

Schedule

Topics	Weeks
Introduction to Data Science: Applications, Roles, and Workflows	1
Data Engineering Fundamentals: Collection, Cleaning, and Transformation	2
Descriptive Analytics: Summary statistics and Visualization	2
Predictive Analytics: Regression and Classification	1.5
Pattern Discovery: Clustering and Dimensionality Reduction	1
Reasoning Under Uncertainty: Probability, Sampling, and Estimation	1.5
Testing Claims with Data: Hypothesis Testing and Statistical vs. Practical Significance	1
Designing and Evaluating Experiments: A/B Testing	2
Working with Non-Tabular Data: Text Analytics	1.5
Responsible Data Science: Privacy, Fairness, Transparency, and Communication	0.5

Your Grade is Based on

Assignments:	15% - 3 assignments
Quizzes:	15% - 4 take 3
	(no makeup for missed quizzes)
Project:	15% - Report and Presentation
Midterm Exam:	20% - Week 7
Final Exam:	35%

Data Science Learning Approach

- **Conceptual Foundations:** Explains the "why" behind each method so students build intuition and can choose the right approach for a real-world problem
  - Example-driven teaching using real datasets to make concepts intuitively clear
- **Statistical Foundations:** Builds the probability, sampling, and inference reasoning that underpins sound analysis and honest interpretation of results
- **Learning by doing:**
  - **Hands-on analysis in Python:** Students acquire, clean, explore, visualize, and model real data using NumPy, Pandas, Matplotlib/Seaborn, and scikit-learn
  - **Integrated data science project:** An end-to-end project reinforces problem formulation, critical thinking, responsible data practices, and communication of results
- **Responsible practice throughout:** Privacy, fairness, transparency, and reproducibility are treated as core skills, not afterthoughts

Quizzes & Assignments

- **Quizzes:**
  - Short (~15 minutes) quizzes on Blackboard
  - Mostly multiple-choice questions covering the main concepts discussed in class
  - They give you a good idea of your level of understanding
- **Assignments:**
  - Three Python notebook assignments covering data wrangling, exploratory analysis and visualization, and inference and modeling
  - You will work with real-world datasets and submit reproducible, well-documented code
  - Each task is graded separately

Exams and Project

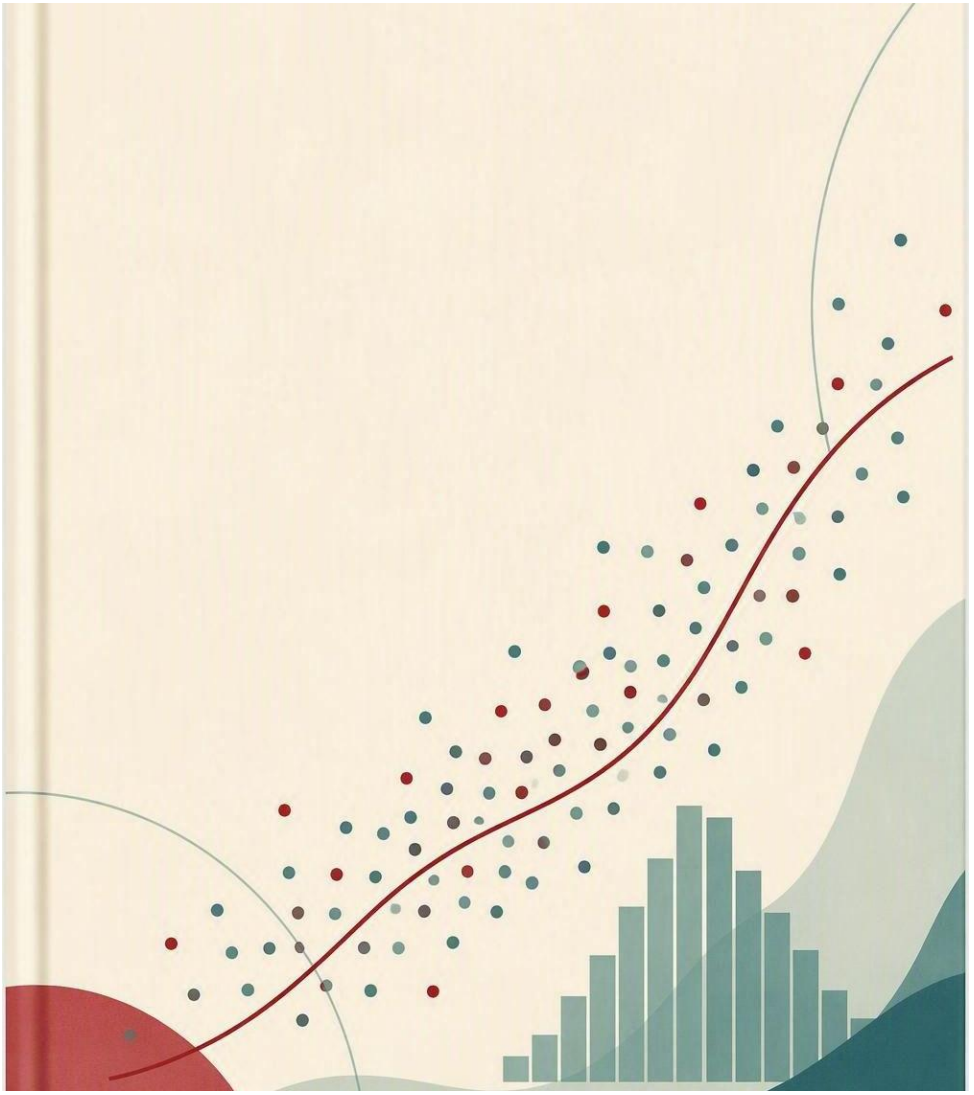
- Exams assess understanding and basic application of data science concepts
  - Midterm Exam in Week 7 (20%) and Final Exam (35%)
- Project: apply the full data science lifecycle to a real problem – you select the dataset and question, then acquire, clean, explore, analyze, and model the data
  - Deliverables are a written report and a presentation of your findings (15%)

Recommended Textbook

Ani Adhikari and John DeNero, **Computational and Inferential Thinking: The Foundations of Data Science** – free online textbook

Available online

Plenty of online resources will be provided



Software we will use

- Python and its data science libraries: NumPy, Pandas, Matplotlib/Seaborn, and scikit-learn
- Jupyter Notebook or Google Colab
- VS Code
- GitHub Desktop
- Blackboard and Microsoft Teams for materials, submissions, and announcements
- Other tools will be communicated to you as we progress



GitHub will be used to deliver Slides, Examples, Assignments, and Project

https://github.com/cmps360f26/cmps360-content

Check it regularly!

Communication

- Post your technical questions to Teams Q&A channel

Do NOT send me by email

- To contact me do not send emails but use **Microsoft Teams** chat

- For **guidance** on technical issues come to office hours NOT by email
- Check Teams regularly for announcements, materials, and deadlines

How to succeed in this course….

- Do your weekly assigned readings
- **Read the slides before you come to the class**
- **Practice a lot – run and modify the notebooks and study as many examples as possible**



  - Understand and extend the examples I provide, as well as those in the textbook and the provided datasets
- **Attend and participate in class**
  - **Many of the exam questions are from the class explanation**
- Do all the assignments and project **yourself.** Actively contribute to your project.
- Seek help when needed and ask questions (and do it EARLY):
  During Lectures & Come to office hours





We learn swimming by swimming and we learn data science by practicing it!

Important Notes

- Attendance… QU attendance policies will be enforced
  - Do not miss classes/labs
- **Start your assignments and project early!!!**
- This is a project-based course and students are expected to learn independently as much as needed in order to complete the course requirements
  - Do not expect me to find/fix your code bugs or debug your notebooks
  - Do not expect me to find and fix your technical issues

  => I can only give you high level suggestions and guidance

No ‘Free Riding’ allowed

- ‘free riders’ (who do not contribute much) => not acceptable and not fair for hardworking students
  - You must actively contribute to your project and do your ultimate best to deliver the best possible results
  - Otherwise you will be asked to do the project alone
  - **Report free-riders early**









Plagiarism / Cheating

- “Getting an unfair academic advantage"
  - Using other people's work as your own
  - Not doing your assignments yourself
- All the code you submit must be your own
    - Only exception: Code I have provided or explicitly authorized
    - **NO** code you have found on the web. **NO** sharing with others.
- **Do your homework and project yourself**
  - Do NOT copy from each other or from the Internet - **I will know it!**
  - You can be picked-up randomly to explain your implementation
  - Cheating will be treated very seriously
- Penalties START with a zero on the assignment, failing the course! and other disciplinary actions as per QU policy

Generative AI Use Policy

- Generative AI may be used as a learning aid, **but not to replace your own work, reasoning, analysis, or original writing**
- **Allowed:** clarify concepts, review syntax, brainstorm approaches, improve writing clarity, and debug or explain code
- **Not allowed:** generating complete solutions, performing the main analysis, or producing assignment, quiz, exam, notebook, report, visualization, or project work for submission

To do before next class

- Install the required software: Python, VS Code, Jupyter Notebook and GitHub Desktop (see announcement on Teams)
- Create your GitHub account **(firstname-quUsername)**
- Decide your project team members and enter them in the spreadsheet on Teams
- Read the first chapter of the textbook
- Prepare any questions you might have



I wish you a fruitful and enjoyable journey!